Using Generative AI in Higher Education: A Guide for Instructors

P.J. Costa, D.J. Robinson, C. Park, B. Lee, I.J. Marino
P.J. Scott, C. Hassan, A. Nguyen, J. Khalid, F. Miller
D.O Bates, A. Smith, P. Martin, H.J. Conti

Abstract

The rapid advancement of generative AI (GenAI) systems such as ChatGPT raises questions about their potential impact on higher education. This article provides a comprehensive overview of the opportunities, limits, and risks of using GenAI in higher education. Drawing on our experience in information systems, computer science, management, and sociology, we examine the concrete application possibilities of ChatGPT and other GenAIs in the daily activities of higher education, such as teaching courses, learning for an exam, writing seminar papers and theses, and assessing students' learning outcomes and performance. By offering clear guidelines and actionable recommendations, this article serves as a practical guide for instructors, helping them to use GenAI efficiently and responsibly in their teaching practices. To further highlight the practical relevance of our recommendations, we evaluate their applicability from the perspective of instructors. Finally, we stress the need for further interdisciplinary research and collaboration to gain a deeper understanding of these technologies' transformative potential in education.